

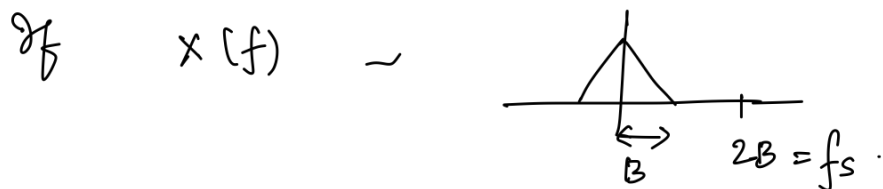
2. Part B.

Problem 1 : (a) $\text{Re}(Y(f)) = \text{Re}(X(f))$

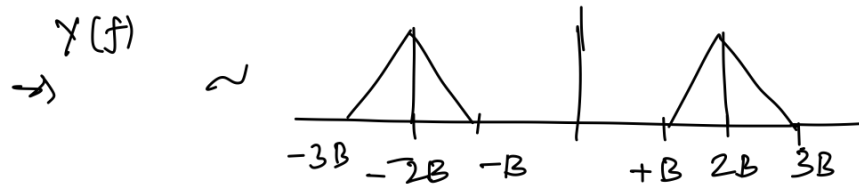
Cannot be determined as imaginary part of $Y(f)$ can be arbitrary.

(b) $y(t) = x(t) \cos 2\pi f_s t$.

$$\Rightarrow Y(f) = X(f) * \left(\frac{\delta(f-f_s) + \delta(f+f_s)}{2} \right) \\ = \frac{X(f-f_s) + X(f+f_s)}{2}$$



$$f_s = 2B$$



$\Rightarrow$ Bandpass signal. (covered in tut).

Sampling freq remains $2B = f_s$.

(c) $y(t) = x(t/2) \Rightarrow Y(f) = 2X(2f)$

bw of $X(2f) = B/2$

$\Rightarrow f_s$ for $y(t) = B$.

(d) $y(t) = x(t-d) \leftrightarrow e^{j2\pi f d} X(f)$.

bw of $y(t) = B \Rightarrow f_s = 2B$.

(e) $y(t) = x(t - \theta(t))$. f_s can be anything as phase varies.

